

Part II V1 in Lamport notation

Thermal Weyl functionals, Mehler bridges, and the exact mean shift

Fresh reconstruction of `part_ii/v1_thermal_representations.tex`

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All finite-dimensional identities in V1 are reconstructed below. The reference measure in the mean-shift formula is the *free periodic Gaussian* measure. The continuum interacting marked-slice identity is not assumed here; it is proved by the V3 reconstruction and consumed only afterward.

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1 Objects and statement

Fix $L, m > 0$, $\varepsilon_N = L/r_N$, the spatial lattice Λ_N with $2r_N$ sites, and

$$\Gamma_N = \{\pi j/L : -r_N \leq j < r_N\}, \quad \gamma_N(k)^2 = m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2).$$

On $L^2(\mathbb{R}^{\Lambda_N}, d\boldsymbol{\varphi})$ let $\Phi_N(x) = \varphi_x$ and $\Pi_N(x) = -i\varepsilon_N^{-1} \partial_{\varphi_x}$. For real q, p set

$$\Phi_N(q) = \varepsilon_N \sum_x q_x \Phi_N(x), \quad \Pi_N(p) = \varepsilon_N \sum_x p_x \Pi_N(x), \quad W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}.$$

Let $H_N^{(0)}$ be the nearest-neighbour oscillator Hamiltonian and

$$K_N = H_N^{(0)} + V_N, \quad V_N = \lambda \varepsilon_N \sum_x :\Phi_N(x)^4:_{c_{N,L}}.$$

The model-sheet module proves that K_N is self-adjoint, bounded below, has compact resolvent, and has a simple ground state.

2 Structured proof

$\langle 1 \rangle$ 1 Derive the exact Schrödinger action of the Weyl operator.

$\langle 2 \rangle$ 1 For $s \in \mathbb{R}$ define

$$(U_s \psi)(\boldsymbol{\varphi}) = e^{is\varepsilon_N q \cdot \boldsymbol{\varphi} + \frac{i}{2} s^2 \varepsilon_N q \cdot p} \psi(\boldsymbol{\varphi} + sp).$$

⟨3⟩ 1 U_s is unitary.

Because: Translation preserves Lebesgue measure and the prefactor has modulus one.

⟨3⟩ 2 $U_{s+s'} = U_s U_{s'}$.

Because: Direct substitution gives the exponent

$$s\varepsilon_N q \cdot \varphi + s'\varepsilon_N q \cdot (\varphi + sp) + \frac{1}{2}(s^2 + s'^2)\varepsilon_N q \cdot p = (s + s')\varepsilon_N q \cdot \varphi + \frac{1}{2}(s + s')^2\varepsilon_N q \cdot p.$$

⟨3⟩ 3 $s \mapsto U_s$ is a strongly continuous unitary group and its generator extends $\Phi_N(q) + \Pi_N(p)$ on C_c^∞ .

Because: Strong continuity first holds on C_c^∞ by dominated convergence and then on all L^2 by density and unitarity. On the core,

$$\left. \frac{d}{ds} U_s \psi \right|_{s=0} = (i\varepsilon_N q \cdot \varphi + p \cdot \nabla) \psi = i(\Phi_N(q) + \Pi_N(p)) \psi.$$

Stone's theorem gives a self-adjoint generator extending the displayed differential operator. Moreover $U_s C_c^\infty \subset C_c^\infty$ for every s ; Nelson's invariant-domain criterion therefore makes C_c^∞ a core for that generator. Thus the differential operator is essentially self-adjoint there and its closure generates U_s .

⟨2⟩ 2 At $s = 1$,

$$(W_N(\varepsilon_N q + ip)\psi)(\varphi) = e^{i\varepsilon_N q \cdot \varphi + \frac{i}{2}\varepsilon_N q \cdot p} \psi(\varphi + p), \quad (1.1)$$

$$W_N(\varepsilon_N q + ip) = e^{\frac{i}{2}\varepsilon_N q \cdot p} e^{i\Phi_N(q)} e^{i\Pi_N(p)}. \quad (1.2)$$

⟨2⟩ 3 This proves ⟨1⟩1. □

⟨1⟩ 2 Prove trace class and the marked-slice kernel identity.

⟨2⟩ 1 K_N has compact resolvent and is bounded below; hence e^{-tK_N} is trace class for $t > 0$.

Because: The model sheet proves $K_N \geq a(-\Delta_\varphi + |\varphi|^2) - b$ in quadratic-form sense for some $a > 0, b < \infty$. The comparison oscillator has compact resolvent and summable heat eigenvalues; the min–max principle yields $\text{Tr } e^{-tK_N} < \infty$.

⟨2⟩ 2 Let $k_t(\mathbf{x}, \mathbf{y})$ be the continuous kernel of e^{-tK_N} . For bounded multiplication f and translation $T_p \psi(x) = \psi(x + p)$,

$$\text{Tr}(e^{-tK_N} M_f T_p) = \int_{\mathbb{R}^N} k_t(\mathbf{x} + p, \mathbf{x}) f(\mathbf{x}) d\mathbf{x}. \quad (2.1)$$

Because: Expand the trace in a real eigenbasis $K_N u_j = E_j u_j$. Its scalar series is absolutely convergent because $M_f T_p$ is bounded. To justify exchanging the series and spatial integration, put $D(x) = e^{-(t/2)K_N}(x, x)$ and use Cauchy–Schwarz twice:

$$\begin{aligned} \sum_j e^{-tE_j} |u_j(x) u_j(x + p)| &\leq \left(\sum_j e^{-tE_j/2} \right) D(x)^{1/2} D(x + p)^{1/2}, \\ \int D(x)^{1/2} D(x + p)^{1/2} dx &\leq \left(\int D(x) dx \right)^{1/2} \left(\int D(x + p) dx \right)^{1/2} = \text{Tr } e^{-(t/2)K_N} < \infty. \end{aligned}$$

Thus Fubini applies. The kernel series is the continuous heat kernel, and the change of variable in the trace gives (2.1) with $f(x)$ (not $f(x + p)$).

⟨2⟩ 3 With $f(x) = e^{i\varepsilon_N q \cdot x}$, (1.1) and (2.1) give

$$\mathrm{Tr}(e^{-tK_N} W_N(\varepsilon_N q + ip)) = e^{\frac{i}{2}\varepsilon_N q \cdot p} \int e^{i\varepsilon_N q \cdot x} k_t(x + p, x) dx. \quad (2.2)$$

⟨2⟩ 4 This proves ⟨1⟩2. □

⟨1⟩ 3 Diagonalize the free system and compute its thermal Weyl functional.

⟨2⟩ 1 Set $X_x = \varepsilon_N^{1/2} \Phi_N(x)$ and $P_x = \varepsilon_N^{1/2} \Pi_N(x)$. The unitary discrete Fourier transform gives

$$H_N^{(0)} = \sum_{k \in \Gamma_N} \gamma_N(k) (a_k^* a_k + \tfrac{1}{2}),$$

$$\tilde{X}(k) = (2\gamma_N(k))^{-1/2} (a_k + a_{-k}^*), \quad \tilde{P}(k) = -i(\gamma_N(k)/2)^{1/2} (a_k - a_{-k}^*). \quad (3.1)$$

Because: Fourier transformation diagonalizes the circulant nearest-neighbour matrix; substitution of (3.1) and $[a_k, a_{k'}^*] = \delta_{kk'}$ cancels the aa and a^*a^* terms and leaves the displayed oscillator sum.

⟨2⟩ 2 For real q, p ,

$$\Phi_N(q) + \Pi_N(p) = \sum_k (\bar{z}_k a_k + z_k a_k^*), \quad z_k = \varepsilon_N^{1/2} \left[\frac{\tilde{q}(k)}{\sqrt{2\gamma_N(k)}} + i \sqrt{\frac{\gamma_N(k)}{2}} \tilde{p}(k) \right]. \quad (3.2)$$

Moreover

$$\sum_k |z_k|^2 = \frac{1}{2(2r_N)} \sum_k \left(\gamma_N(k)^{-1} |\hat{q}(k)|^2 + \gamma_N(k) |\hat{p}(k)|^2 \right). \quad (3.3)$$

Because: Parseval gives (3.2). In $|z_k|^2$ the cross term is $\mathrm{Im}(\tilde{q}(k) \overline{\tilde{p}(k)})$; it is odd under $k \mapsto -k$ and cancels in the sum. The relation between the two Fourier normalizations gives (3.3).

⟨2⟩ 3 For one oscillator $H = \gamma(a^*a + 1/2)$ and $D(z) = e^{za^* - \bar{z}a}$,

$$\frac{\mathrm{Tr}(e^{-tH} D(z))}{\mathrm{Tr} e^{-tH}} = \exp[-\tfrac{1}{2}|z|^2 \coth(t\gamma/2)]. \quad (3.4)$$

Because: BCH gives $D(z) = e^{-|z|^2/2} e^{za^*} e^{-\bar{z}a}$. In the number basis,

$$\langle n | e^{za^*} e^{-\bar{z}a} | n \rangle = L_n(|z|^2), \quad \sum_{n \geq 0} w^n L_n(x) = \frac{e^{-xw/(1-w)}}{1-w}.$$

The first identity follows by expanding both exponentials and matching the number change; the second follows from $\sum_{n \geq j} \binom{n}{j} w^n = w^j (1-w)^{-j-1}$. Put $w = e^{-t\gamma}$ and use $(1+w)/(1-w) = \coth(t\gamma/2)$.

⟨2⟩ 4 The finite tensor-product trace therefore equals

$$F_N^{(0)}(t; \varepsilon_N q + ip) = \exp[-\tfrac{1}{4} Q_N^{(t)}(q, p)], \quad (3.5)$$

$$Q_N^{(t)}(q, p) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} \coth(t\gamma_N(k)/2) \left(\gamma_N(k)^{-1} |\hat{q}(k)|^2 + \gamma_N(k) |\hat{p}(k)|^2 \right). \quad (3.6)$$

Also $\mathrm{Tr} e^{-tH_N^{(0)}} = \prod_k [2 \sinh(t\gamma_N(k)/2)]^{-1}$.

Because: Apply (3.4) to each commuting oscillator and insert (3.3); the zero-point geometric sum is $[2 \sinh(t\gamma/2)]^{-1}$.

⟨2⟩ 5 This proves ⟨1⟩3. □

⟨1⟩ 4 Verify the Mehler kernel and the unique harmonic jump interpolant.

⟨2⟩ 1 For $H_\omega = \frac{1}{2}(-\partial_x^2 + \omega^2 x^2)$, set

$$k_\Delta^\omega(x, y) = \left(\frac{\omega}{2\pi \sinh(\omega\Delta)} \right)^{1/2} \exp \left[-\frac{\omega \coth(\omega\Delta)}{2}(x^2 + y^2) + \frac{\omega xy}{\sinh(\omega\Delta)} \right]. \quad (4.1)$$

⟨3⟩ 1 $\partial_\Delta k_\Delta^\omega = -H_\omega k_\Delta^\omega$.

Because: Differentiate the prefactor and exponent, using $(\omega \coth \omega\Delta)' = -\omega^2 \operatorname{csch}^2(\omega\Delta)$ and $(\omega \operatorname{csch} \omega\Delta)' = -\omega^2 \coth(\omega\Delta) \operatorname{csch}(\omega\Delta)$; the coefficients of $1, x^2, xy, y^2$ agree separately.

⟨3⟩ 2 $k_\Delta^\omega \rightarrow \delta_{x=y}$ as $\Delta \downarrow 0$.

Because: In variables $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$, the quadratic form has coefficients $\omega \tanh(\omega\Delta/2)$ and $\omega \coth(\omega\Delta/2)$; the latter tends to infinity with the normalized Gaussian mass, while the former tends to zero.

⟨3⟩ 3 Thus (4.1) is the heat kernel.

Because: For compactly supported smooth initial data, both convolution with (4.1) and $e^{-\Delta H_\omega}$ solve the same L^2 Cauchy problem. Their difference v obeys $d\|v\|_2^2/d\Delta = -2\langle v, H_\omega v \rangle \leq 0$ and starts at zero.

⟨2⟩ 2 For a prescribed jump ρ , the boundary-value problem

$$-\ddot{H} + \omega^2 H = 0, \quad H(t) - H(0) = \rho, \quad \dot{H}(t) = \dot{H}(0)$$

has the unique solution

$$H(s) = \frac{\rho \sinh(\omega(s - t/2))}{2 \sinh(\omega t/2)}. \quad (4.2)$$

Because: The general solution is $A \cosh(\omega s) + B \sinh(\omega s)$. Solving the two linear boundary equations gives $A = -\rho/2$ and $B = (\rho/2) \coth(\omega t/2)$, which is (4.2) by the subtraction formula. The homogeneous system has only $A = B = 0$.

⟨2⟩ 3 The interpolant satisfies

$$|H(s)| \leq |\rho|/2, \quad S(H) := \frac{1}{2} \int_0^t (\dot{H}^2 + \omega^2 H^2) ds = \frac{\omega \rho^2}{4} \coth(\omega t/2). \quad (4.3)$$

Because: $|s - t/2| \leq t/2$ and $|\sinh|$ is increasing on $[0, \infty)$. Integration by parts and the ODE give $S(H) = \frac{1}{2}[H\dot{H}]_0^t = \frac{1}{2}\rho\dot{H}(0)$; differentiating (4.2) at 0 gives the last expression.

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Prove the exact discrete Gaussian shift identity.

Fix the equal partition $s_i = i\Delta$, $\Delta = t/M$, and let $H_i = H(s_i)$. Define the periodic Mehler-chain integral

$$J_0[f_0, \dots, f_{M-1}] = \int \prod_{i=0}^{M-1} k_\Delta^\omega(\eta_{i+1}, \eta_i) f_i(\eta_i) \prod_i d\eta_i, \quad \eta_M = \eta_0,$$

and the jump chain J_ρ by replacing the closing condition by $x_M = x_0 + \rho$.

⟨2⟩ 1 Under $x_i = \eta_i + H_i$, every interior Mehler exponent has zero linear term in η_i .

Because: Apart from its normalization, the negative logarithm of one kernel is

$$Q_\Delta(y, x) = \frac{\omega}{2 \sinh(\omega \Delta)} [\cosh(\omega \Delta)(x^2 + y^2) - 2xy].$$

For $1 \leq i < M$, the coefficient of η_i in $\sum_i Q_\Delta(\eta_{i+1} + H_{i+1}, \eta_i + H_i)$ is proportional to

$$2 \cosh(\omega \Delta) H_i - H_{i-1} - H_{i+1} = 0,$$

the exact three-term propagation law of every solution of $-\ddot{H} + \omega^2 H = 0$. The combined coefficient of the periodic variable $\eta_0 = \eta_M$ is the difference of the endpoint classical momenta and vanishes because $\dot{H}(t) = \dot{H}(0)$. Hence all linear terms, including the closing one, vanish.

$\langle 2 \rangle 2$ The constant part of the exponent is $S(H)$, independent of the mesh.

Because: $Q_\Delta(H_{i+1}, H_i)$ is exactly the classical action $\frac{1}{2} \int_{s_i}^{s_{i+1}} (\dot{H}^2 + \omega^2 H^2) ds$ of the unique harmonic path joining the two endpoints. The restriction of (4.2) is that path. Hence

$$\sum_i Q_\Delta(H_{i+1}, H_i) = \frac{1}{2} \int_0^t (\dot{H}^2 + \omega^2 H^2) ds = \frac{1}{2} [H \dot{H}]_0^t = S(H).$$

$\langle 2 \rangle 3$ The endpoint conditions transform correctly: $x_M = x_0 + \rho$ becomes $\eta_M = \eta_0$ because $H(t) - H(0) = \rho$.

$\langle 2 \rangle 4$ Therefore

$$J_\rho[f_0, \dots, f_{M-1}] = e^{-S(H)} J_0[f_0(\cdot + H_0), \dots, f_{M-1}(\cdot + H_{M-1})]. \quad (5.1)$$

The multimode identity is the product of (5.1).

$\langle 2 \rangle 5$ This proves $\langle 1 \rangle 5$. □

$\langle 1 \rangle 6$ Derive the interacting marked-slice mean-shift formula.

$\langle 2 \rangle 1$ The harmonic-reference FKN formula gives

$$k_t^K(x, y) = k_t^{(0)}(x, y) \mathbb{E}_{x,y}^{t,0} \exp \left[- \int_0^t V_N(B_s) ds \right]. \quad (6.1)$$

Because: For bounded truncations of V_N this is the Trotter product formula with the Mehler kernel. The Wick quartic is continuous and bounded below; truncate it from above, apply monotone/dominated convergence to the positive kernels, and use the closed lower-bounded form limit. This proves (6.1) without importing an interacting scaling limit.

$\langle 2 \rangle 2$ Insert (6.1) into (2.2) and approximate the time integral by Riemann sums.

Because: Each bridge path is continuous and V_N is continuous, so the Riemann sums converge pointwise. If $V_N \geq -c_N$, every exponential is bounded by e^{tc_N} ; the remaining Mehler chain has finite mass. Dominated convergence applies.

$\langle 2 \rangle 3$ Apply (5.1) mode by mode with jump $\rho_k = \varepsilon_N^{1/2} \tilde{p}(k)$.

Because: The position translation in (1.1) is p , hence in the rescaled mode coordinate it is exactly ρ_k . The product Jacobian is one.

$\langle 2 \rangle 4$ The phase from (1.1) cancels the value of the marked position source on $H(0) = -p/2$.

Because: The inserted factor becomes

$$e^{\frac{i}{2}\varepsilon_N q \cdot p} e^{i\varepsilon_N q \cdot (\eta(0) - p/2)} = e^{i\varepsilon_N q \cdot \eta(0)}.$$

⟨2⟩ 5 Let $\mu_{N,t}^{\text{per}}$ be the free periodic Gaussian field with mode covariance

$$c_\omega(s) = \frac{\cosh(\omega(t/2 - |s|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}. \quad (6.2)$$

Then

$$\frac{\text{Tr}(e^{-tK_N} W_N(\varepsilon_N q + ip))}{\text{Tr} e^{-tK_N}} = e^{-S_N(H_p)} \frac{\mathbb{E}_{\mu_{N,t}^{\text{per}}}[e^{i\varepsilon_N q \cdot \eta(0)} e^{-\mathcal{U}_N(\eta + H_p)}]}{\mathbb{E}_{\mu_{N,t}^{\text{per}}}[e^{-\mathcal{U}_N(\eta)}]}. \quad (6.3)$$

Because: The periodic Mehler chain is centered Gaussian and its two-point function is (6.2), obtained either by inverting $-\partial_s^2 + \omega^2$ on the time circle or by the oscillator trace. Normalize the periodic chain by the zero-source partition function. Steps ⟨2⟩2–⟨2⟩4 give exactly the numerator; $q = p = 0$ gives the denominator.

⟨2⟩ 6 The shifted Wick quartic is exactly

$$:(\eta + H)^4:_c - :\eta^4:_c = 4H:\eta^3:_c + 6H^2:\eta^2:_c + 4H^3\eta + H^4. \quad (6.4)$$

Because: Expand $(\eta + H)^4 - 6c(\eta + H)^2 + 3c^2$ and use $\eta^3 = :\eta^3:_c + 3c\eta$ and $\eta^2 = :\eta^2:_c + c$; all additional c terms cancel.

⟨2⟩ 7 This proves ⟨1⟩6. □

⟨1⟩ 7 Construct the continuum free thermal functional.

Let $\gamma = (-\partial_x^2 + m^2)^{1/2}$ on $\mathbb{T}_L$ and

$$h_{\text{ct}} = H^{-1/2}(\mathbb{T}_L) \oplus iH^{1/2}(\mathbb{T}_L).$$

⟨2⟩ 1 For $\zeta = q + ip \in h_{\text{ct}}$, the symplectic form $\text{Im} \int \bar{\zeta} \zeta'$ is finite and defines the continuum Weyl operator $W_{\text{ct}}(\zeta)$.

Because: Sobolev duality pairs $H^{-1/2}$ with $H^{1/2}$. The covariance norm $\langle q, \gamma^{-1}q \rangle + \langle p, \gamma p \rangle$ is finite by definition.

⟨2⟩ 2 $e^{-t d\Gamma(\gamma)}$ is trace class and

$$\text{Tr} e^{-t d\Gamma(\gamma)} = \prod_{k \in (\pi/L)\mathbb{Z}} (1 - e^{-t\gamma(k)})^{-1}.$$

Because: The occupation basis gives the product of geometric series. It converges because $\sum_k e^{-t\gamma(k)} < \infty$.

⟨2⟩ 3 For square-summable displacement amplitudes, absolute convergence permits factorization of the trace into one-mode traces. Hence

$$F_\infty^{(0)}(t; q + ip) = \exp \left[-\frac{1}{4} \sum_k \coth(t\gamma(k)/2) \left(\gamma(k)^{-1} |\hat{q}(k)|^2 + \gamma(k) |\hat{p}(k)|^2 \right) \right], \quad (7.1)$$

with the Fourier normalization fixed by the model sheet.

Because: $|\langle n, D(z)n \rangle| \leq 1$ dominates the trace by the partition function; Fubini applies, and (3.4) evaluates every factor.

⟨2⟩ 4 The D4 image of every fixed coarse symbol belongs to h_{ct} .

Because: Its Fourier coefficients contain $P_\infty(\varepsilon_M k)$ with $|P_\infty(\ell)|^2 \leq C_D(1 + |\ell|)^{-2-\eta}$, $\eta > 0$. The momentum sum is bounded by $\sum_k |k|(1 + \varepsilon_M |k|)^{-2-\eta} < \infty$, and the position sum is easier.

⟨2⟩ 5 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the free thermal fixed-coarse rate uniformly for $t \geq t_0 > 0$.

⟨2⟩ 1 For $w_t(\gamma) = \coth(t\gamma/2)$ and $\bar{w} = \coth(t_0 m/2)$,

$$\begin{aligned} 1 &\leq w_t(\gamma) \leq \bar{w}, \\ |(\gamma w_t(\gamma))'| &\leq \bar{w} + s_*, \\ |(\gamma^{-1} w_t(\gamma))'| &\leq (\bar{w} + s_*)m^{-2}, \end{aligned}$$

where $s_* = \sup_{x \geq t_0 m/2} x / \sinh^2 x < \infty$.

Because: Differentiate $\coth(t\gamma/2)$ and use $\gamma \geq m$.

⟨2⟩ 2 On low momenta, split each weighted multiplier difference into a dispersion term and a D4-filter term.

Because: For the momentum multiplier,

$$|\gamma_N w_t(\gamma_N)|P_r|^2 - \gamma w_t(\gamma)|P_\infty|^2| \leq |\gamma_N w_t(\gamma_N) - \gamma w_t(\gamma)||P_r|^2 + \gamma w_t(\gamma)||P_r|^2 - |P_\infty|^2|.$$

The first term is at most $(\bar{w} + s_*)\varepsilon_N^2 k^4 / (24m)$ times $|P_r|^2$; the position multiplier is the same with the derivative bound $m^{-2}(\bar{w} + s_*)$. The second terms use the explicit finite-product filter tail.

⟨2⟩ 3 On high momenta, multiply the vacuum D4 envelope estimate by at most $\bar{w}$.

Because: $1 \leq w_t \leq \bar{w}$ uniformly in $t \geq t_0$; no derivative is used in the tail.

⟨2⟩ 4 Choose the same split exponent as the free-rate module, $2 - 5\alpha = \alpha\eta = \theta = 2\eta/(5 + \eta)$. Then

$$|Q_{M+r}^{(t)}(\xi) - Q_\infty^{(t)}(\xi)| \leq C_{M,\xi,m,L,t_0} 2^{-\theta r}, \quad t \in [t_0, \infty]. \quad (8.1)$$

Because: The low sum is $O(4^{-r}(1 + J_r)^5)$ and the high tail is $O(J_r^{-\eta})$. Put $J_r \asymp 2^{\alpha r}$; the two exponents become equal. Every sum and counting constant is the explicit one proved in the free-rate module.

⟨2⟩ 5 The characteristic functionals obey the same rate up to a factor $1/4$.

Because: For $x, y \geq 0$, $|e^{-x/4} - e^{-y/4}| \leq |x - y|/4$.

⟨2⟩ 6 This proves ⟨1⟩8. □

3 V1 conclusion and dependency interface

The proved outputs are the exact lattice Weyl phase, trace-kernel formula, free thermal closed forms, Mehler/jump shift, interacting finite-lattice representation (6.3), and uniform free fixed-coarse rate (8.1). Formula (6.3) uses only a finite-dimensional FKN argument. The analogous continuum interacting formula requires the continuum Hamiltonian/periodic FKN construction; the V3 module proves it before V4 uses it.